



Google Developer Group
Dubai

Introduction To ADK 2.0

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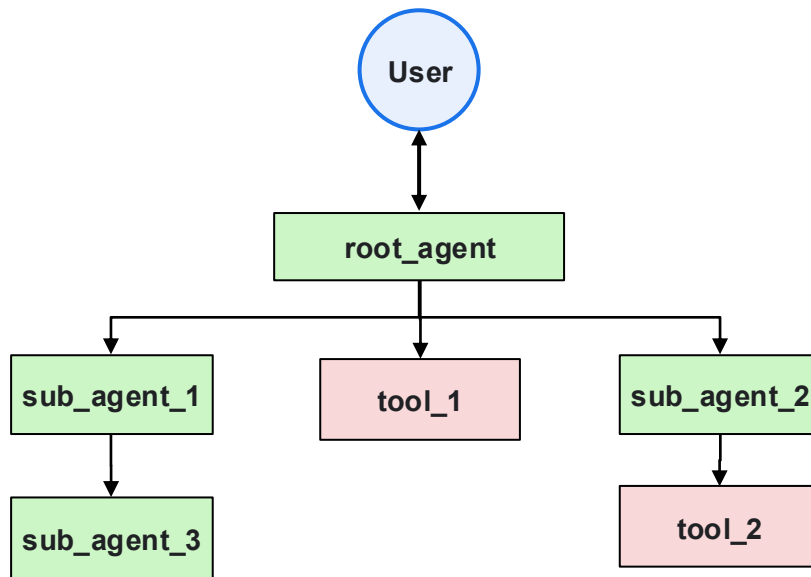


Build  with AI

ADK 1.x - Mental Model



- One **root_agent** talks to the user
- **sub_agents** = specialists (intake, legal, security, manager)
- Coordinator **transfers** via prompts when the LLM decides
- Pipeline order is **implicit**, not declared
- Heavily relies on **prompting** and instruction following



"Route when the model thinks it's time"

ADK 1.x - What Worked

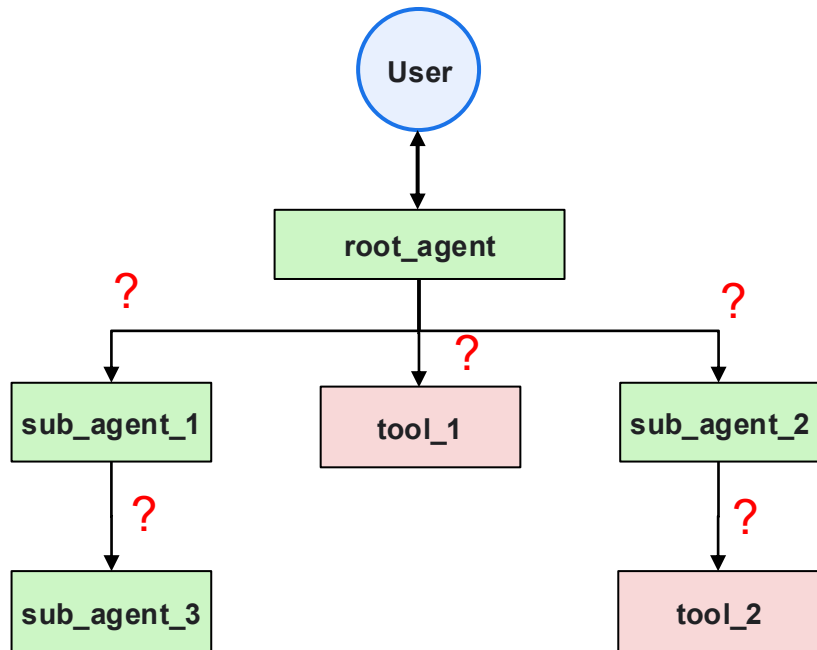


- Fast conversational multi-agent prototypes
- Non-deterministic flow
 - User may ask anything on topic and agent can “figure it out”
- sub_agents + tools still an option in 2.0 (collaborative path)
- Strong fit: delegation-first UX
 - “ask my coordinator”

ADK 1.x - Push To Upgrade



- No pipeline diagram:
 - Less deterministic
 - Hard to review in UI or code review
- Procedure in prompts:
 - Steps can be skipped or reordered
- Parallelism unclear:
 - “Run both tools in parallel” is hope, not structure
- Human-In-The-Loop:
 - Possible, but wiring is ad-hoc



"Route IF the model thinks it's time"

ADK 2.0 - Introducing 3 New Paradigms

Graph Workflows

- Deterministic Pipelines
- Declare flow in edges and use Event routing for branches
- Stable, teachable structure
- Parallel fan-out via tuples
- Explicit state hydration
- Best for clear SOPs

Dynamic Workflows

- Algorithmic Control
- Implement control flow in Python inside `@node` orchestrators
- Loop-heavy logic
- Runtime-shaped branching
- Standard `async/await`
- Automatic checkpointing

Collaborative Workflows

- Agentic Delegation
- Coordinator delegates to specialized sub-agents via tools
- Multi-agent product shapes
- Native mode="task"
- Isolated session branches
- Human-in-the-loop (HITL)

ADK 1.x vs 2.0 at a glance

	ADK 1.x	ADK 2.0
Control	Prompts / transfer	Workflow + Event(route=...) or Python orchestrator
Structure	Hidden in chat	Declared edges (visible in Web UI for graph apps)
Parallel	Implicit	(agent1, agent2) or asyncio.gather
HITL	Custom patterns	require_confirmation tools · RequestInput

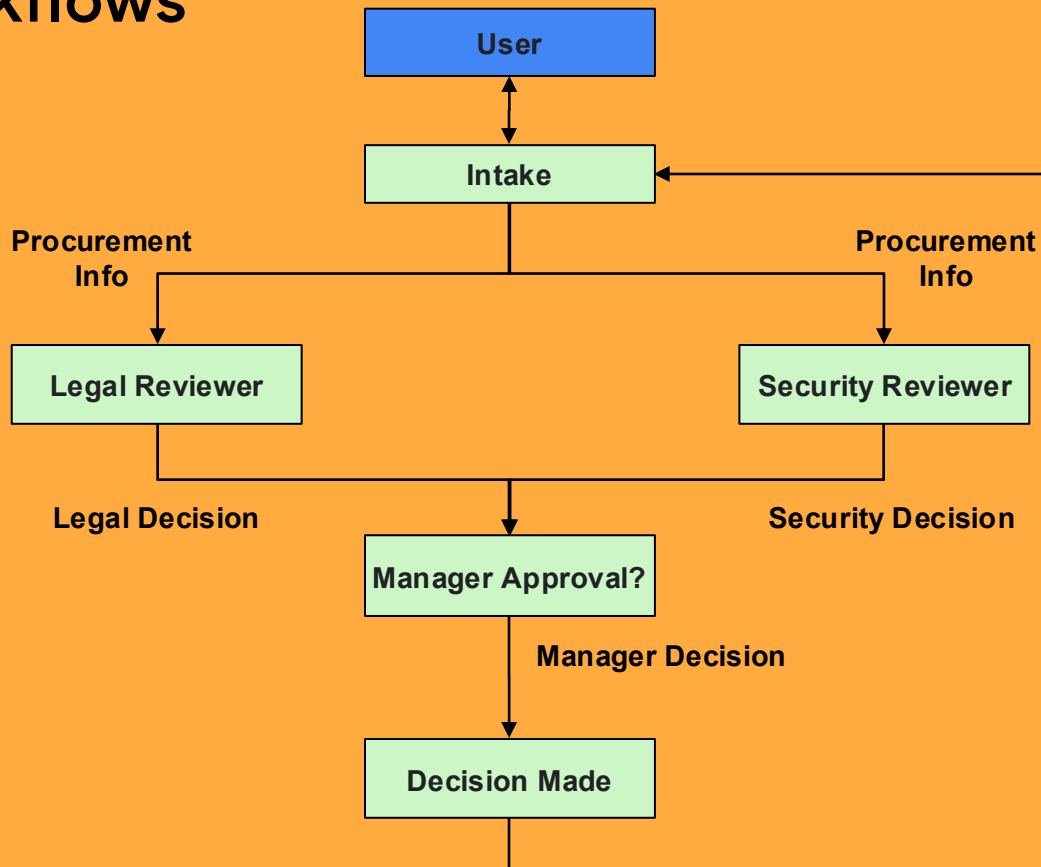
Case Study: AI Procurement Agent

Problem Statement & Requirements

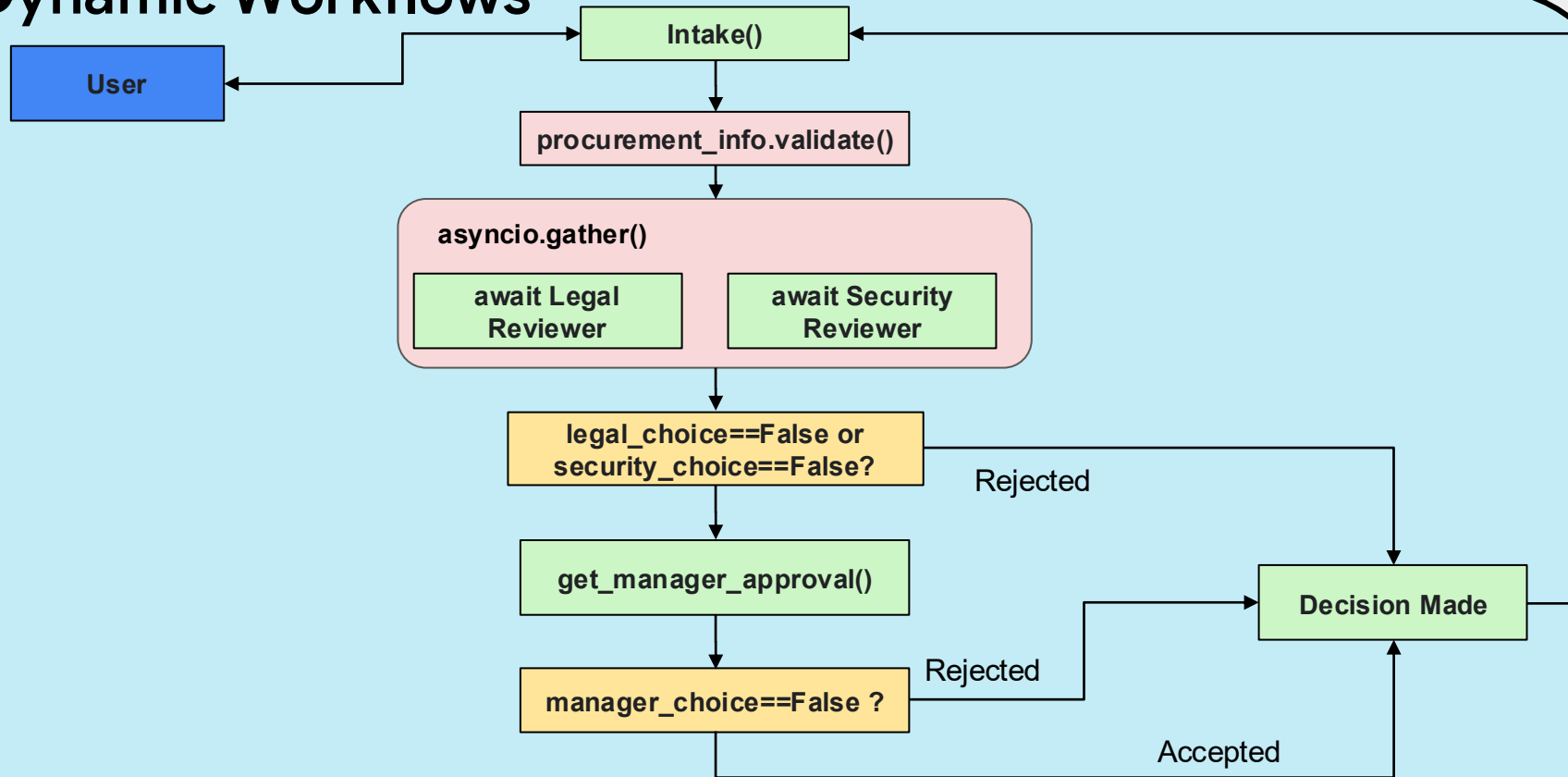
- Automate manual software license and subscription requests
- Gather annual cost in AED and business justification from users
- Execute mandatory security and legal compliance checks
- Auto-approve low-value subscriptions (<500 AED)
- Trigger manager workflow for high-value procurement



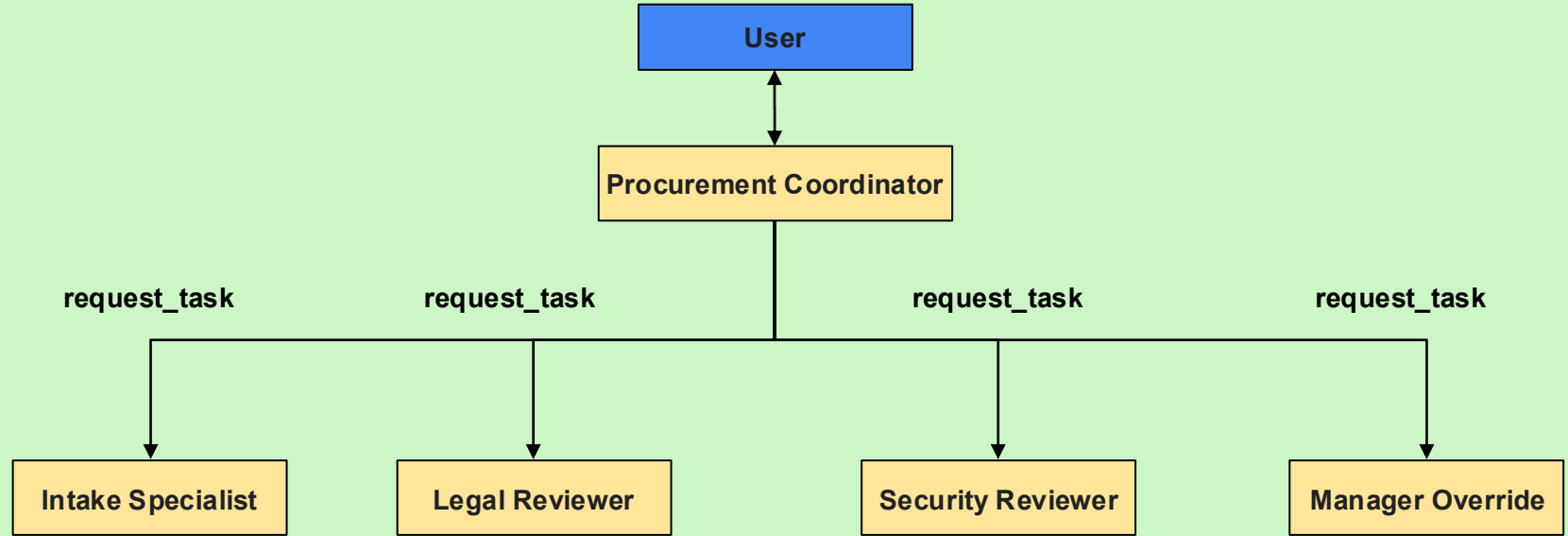
Graph Workflows



Dynamic Workflows



Collaborative Workflows



Code With Me!



Github code - Clone locally



github.com/ro1406/adk-2.0-tutorial

AI Studio - Create Free API Key



<https://aistudio.google.com/api-keys>





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Thank You!



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Thank You!



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